



loomberg Data Scientist Case Study

Background

You are provided with a sample dataset of U.S. flight operations from the Bureau of Transportation Statistics (BTS). This dataset contains information on flight schedules, actual departure/arrival times, and delays.

Objectives

- Work with real-world datasets
- Perform exploratory data analysis (EDA)
- Apply time series or modeling techniques
- Communicate insights clearly

Tasks

1. Data Acquisition

Use the provided sample dataset as a starting point. Retrieve additional relevant data from BTS (<https://www.transtats.bts.gov/>) or other external sources. Clearly document what data you retrieved and how you retrieved it.

2. Exploratory Data Analysis (Required)

Perform EDA to understand delay distributions, airport/airline differences, and temporal patterns. Highlight anomalies and data quality issues.

3. Choose ONE of the following

Option A — Time Series Analysis:

- Define a meaningful metric (e.g., delays, delay rate, flight volume)
- Choose a time granularity (daily, weekly, etc.) and justify it
- Explore patterns: trends, seasonality, variability
- Identify anomalies and give possible explanations
- (Optional) Build a simple forecast and evaluate it
- Summarize key insights and real-world implications

Option B — Delay Analysis & Modeling

- Define the prediction task (regression or classification)
- Select and engineer relevant features (time, airport, airline, etc.)
- Train at least one model and describe approach
- Evaluate model performance with appropriate metrics
- Identify and interpret key drivers of delays
- Summarize insights and connect to real-world factors

4. Additional Insights

Share any additional findings. You are encouraged to enrich your analysis with external datasets (e.g., weather, airport information, holidays).

Deliverables

Please generate the report within a Jupyter Notebook and upload it to your GitHub. Please share the GitHub repository link with us. The case study should include a report summarizing your findings and recommendations, as well as any code, visualizations, or other materials that were used in your analysis. The report should be structured in a way that is easy to understand for both technical and non-technical audiences and should clearly communicate the key insights and recommendations that were identified during the analysis.

Thank you and Good Luck!